

SYED SALIK ALVI

Machine Learning & Generative AI Engineer

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SUMMARY

Generative AI engineer focused on building and shipping agentic systems with LangChain, LangGraph, and LlamaIndex. Comfortable across RAG pipelines, multi node agent orchestration, tool calling, and human in the loop design. Owns features end to end and translates technical work for non technical stakeholders.

EDUCATION

Sir Syed University of Engineering & Technology

Bachelor of Science in Computer Science | GPA: 3.25 / 4.00

Oct 2022 – Jul 2026

Karachi, Pakistan

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Operating Systems, Database Systems, Software Engineering, Computer Networks, Software Quality Assurance, Artificial Intelligence.

EXPERIENCE

AI / Machine Learning Intern

Dreams Network and Technology

Aug 2025 – Jan 2026

Karachi, Pakistan

- Built and deployed AI driven automation pipelines that streamlined operational workflows across teams and removed repetitive data entry.
- Integrated ML models into a sales forecasting pipeline using Python, Scikit learn, and SQL, raising accuracy through feature engineering and model selection.
- Built reusable OOP based pipeline modules that cut integration time for new data sources from days to hours.

Web Development Intern

The Tech Turbine

Aug 2023 – Dec 2023

Karachi, Pakistan

- Built responsive frontend features in JavaScript and React with component based design, delivering reusable UI components across multiple client projects.
- Refactored the REST API consumption layer with senior developers, improving response handling and cutting duplicate network calls.
- Worked in Agile sprints, code reviews, and Git based collaboration on a codebase shared by 5 developers.

PROJECTS

Text to SQL Agent | Next.js, FastAPI, LangGraph, SQLAlchemy, Groq

[Codebase](#) · [Live App](#)

- Built a full stack natural language to SQL agent on a LangGraph state machine (inspect schema, generate, validate, execute, format) using Qwen3-32B via Groq, querying SQLite, PostgreSQL, and MySQL through one SQLAlchemy interface.
- Added three read only safety layers (prompt constraints, a SELECT only validator, a read only DB connection) plus self correction that retries failed queries up to three times, streaming SQL and results over Server Sent Events.

Real Time NBA Analytics Dashboard | Python, XGBoost, LightGBM, Streamlit, Plotly

[Codebase](#)

- Built an end to end ML system that ingests live NBA data and engineers 12 predictive features across team form, matchup history, and game context.
- Trained an XGBoost and LightGBM ensemble for game outcome prediction and shipped it on a Streamlit dashboard with auto refresh and interactive Plotly charts.

Research Agent | Next.js, FastAPI, LangGraph, LangSmith, Groq, Pinecone

[Codebase](#) · [Live App](#)

- Shipped a full stack agentic RAG assistant with a Next.js/Vercel frontend and a containerized FastAPI backend on Hugging Face Spaces, on an 11 node LangGraph pipeline (retrieve, evaluate, search, human review, generate) grounding answers in PDFs via Pinecone retrieval and live web data via Tavily.
- Streamed LLM tokens over Server Sent Events for fast first token output and kept it on free tier infrastructure within Groq's 6,000 TPM limit through context trimming, output caps, and rate limit backoff.

TECHNICAL SKILLS

Languages: Python, C#, JavaScript, SQL, HTML, CSS

Frameworks & Web: .NET, React.js, Next.js, Node.js, FastAPI, REST APIs, Tailwind CSS

Databases & Cloud: Pinecone, ChromaDB, MySQL, SQL Server, Docker, Hugging Face Spaces, Vercel, AWS

GenAI & Agents: LangChain, LangGraph, LangSmith, LlamaIndex, RAG pipelines, Multi agent orchestration, Prompt engineering, Tool / Function calling, Embeddings & semantic search, Model Context Protocol (MCP)

AI / ML & Data: Scikit learn, XGBoost, LightGBM, YOLO (v8 to v11), OpenCV, NumPy, Pandas, Matplotlib, Plotly, Power BI

Concepts: OOP, Data Structures & Algorithms, System Design, RAG, Agent evaluation & observability, Relational Databases, Agile / Scrum, Test Driven Development, Software Quality Assurance

LEADERSHIP & EXTRACURRICULAR

Director of Guest Relations | SSUET Media & Events Council

Nov 2025 – Jan 2026

- Led a team of 12 volunteers coordinating VIP hospitality and protocol logistics for flagship events with 500+ guests.

Corporate Lead | SSUET University Events

Nov 2024 – Jan 2025

- Secured corporate sponsorships for flagship university events, owning outreach, pitching, and negotiation with 20+ companies.